

Design and Planning of a Grid-Connected Hybrid Solar PV System for a Residential House in Patiala, Punjab



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1. Introduction

Rooftop solar photovoltaic (PV) generation has become one of the most practical ways for an Indian household to cut its electricity bill and reduce dependence on the utility. In Punjab, PSPCL allows rooftop systems to be connected under net metering, where units exported to the grid are banked against units imported and settled on an annual basis. Adding a battery to such a system makes it a hybrid system that can also keep essential loads running when the grid fails.

This assignment presents the design and planning of a grid-connected hybrid rooftop solar PV system for a five-member household in a two-storey house in Patiala, Punjab. The consumer can use electricity from three sources: the solar PV array, the battery bank, and the utility grid, depending on solar availability, battery state and load demand. The system is designed to perform the following functions:

- Supply household loads directly from the PV array during daylight hours.
- Charge the battery bank when PV output exceeds the instantaneous load.
- Draw power from the utility grid when neither PV nor the battery can meet the load.
- Export surplus PV energy to the grid through net metering once the battery is charged.
- Supply the essential loads from the battery (and PV, when available) for up to seven days of grid outage.

2. Objective and Design Basis

The objective is to design a grid-connected hybrid rooftop PV system for the house described below, meeting the household energy demand and providing seven days of battery backup for essential loads, while remaining synchronised with the PSPCL grid. The design covers the following tasks:

- Estimate the household load, and calculate daily, monthly and annual energy consumption.
- Assess the available rooftop area and its suitability for PV.
- Size the PV array and select 550 W monocrystalline TOPCon modules.
- Design the string configuration and verify it against the inverter limits at extreme temperatures.
- Size the hybrid inverter and the LiFePO₄ battery bank for 7-day backup of essential loads.
- Select cables and protection devices, and explain grid synchronisation, anti-islanding and net metering.
- Estimate the project cost, annual savings and payback period, and summarise the engineering verdict.

Table 1: Design basis and assumptions

Parameter	Value adopted
Location	Patiala, Punjab, India (30.34° N, 76.39° E)
Household	Five members, two-storey house, three split air conditioners
Backup requirement (assignment brief)	7 days for essential loads
PV module (assignment brief)	550 W monocrystalline half-cut TOPCon
Utility supply	PSPCL, 415 V three-phase, 50 Hz, net metering with annual settlement
Tariff basis	PSERC FY 2026-27 domestic tariff: ₹3.85/kWh up to 300 units per month and ₹7.05/kWh above 300 units (energy charge only)
Battery design factors	LiFePO ₄ , 90% usable depth of discharge, 92% discharge-path efficiency, 90% ageing and temperature derating

Parameter	Value adopted
Design temperatures	-5 °C minimum ambient (string Voc check); 45 °C ambient and 70 °C cell temperature (string Vmp check)

3. Site Details and Rooftop Area

3.1 Plot Area and Roof Area

The house stands on a 10-marla plot. In Punjab, 1 marla = 272.25 ft² = 25.29 m². The ground coverage is assumed to be 60%, and the roof footprint is taken equal to the ground-floor footprint.

Plot area = 10 × 25.29 = 252.9 m² (about 303 sq yd, or 2,723 ft²)

Roof footprint = Plot area × Ground coverage = 252.9 × 0.60 = 151.8 m² (about 1,634 ft²)

The roof is taken as a rectangle of 15.0 m (east-west) × 10.12 m (north-south).

3.2 Usable Rooftop Area

Not all of the roof can carry PV modules. A 0.6 m strip along the parapet is kept clear to avoid parapet shading and to allow access, and several fixed structures take up further space. Table 2 builds up the usable area step by step.

Table 2: Usable rooftop area

Item	Area (m ²)
Gross roof area	151.8
Less: 0.6 m perimeter setback (parapet shadow and access)	-28.7
Inner roof area (13.8 m × 8.92 m)	123.1
Less: Staircase head-room (mumty)	-12.0
Less: Water-tank platform (2 × 1,000 L)	-6.0
Less: Shadow buffer around mumty and tanks	-8.0
Less: Ventilation shafts, solar water heater, dish/antenna	-3.0
Calculated usable area	94.1
Selected usable rooftop area for design	94

4. Load Assessment

4.1 Daily Load on a Peak-Summer Day

The household load is estimated for a peak-summer day, when the air conditioners run and the demand is highest. The air conditioners are modern 1.5-ton inverter splits with an average running input of 1.1 kW, used for a combined 18 hours per day across three bedrooms.

Table 3: Daily load table (peak-summer day)

Load	Quantity	Power (kW)	Hours/day	Energy (kWh/day)
Air conditioners (1.5-ton inverter split)	3 units	1.10	6.0	19.80
Ceiling fans	6 fans	0.075	10.0	4.50
Refrigerator (300 L frost-free)	1 unit	0.20	9.0	1.80

Load	Quantity	Power (kW)	Hours/day	Energy (kWh/day)
Water pump	1 unit	0.75	1.2	0.90
LED lighting	20 lamps	0.012	6.0	1.44
TV and set-top box	2 sets	0.10	4.0	0.80
Washing machine	1 unit	0.50	1.0	0.50
Kitchen (microwave, mixer, induction)	Lumped	1.00	1.0	1.00
Wi-Fi router, laptops, chargers	Lumped	0.25	8.0	2.00
Total daily energy demand				32.74

Air conditioners: $3 \times 1.10 \text{ kW} \times 6 \text{ h} = 19.80 \text{ kWh/day}$

Ceiling fans: $6 \times 0.075 \text{ kW} \times 10 \text{ h} = 4.50 \text{ kWh/day}$

Total = $19.80 + 4.50 + 1.80 + 0.90 + 1.44 + 0.80 + 0.50 + 1.00 + 2.00 = 32.74 \text{ kWh/day}$

4.2 Seasonal Consumption and Annual Energy

Multiplying a peak-summer day by 365 would overstate the annual demand, because the air conditioners are needed for only part of the year and a geyser or room heater is needed in winter. The annual demand is therefore built month by month: the non-air-conditioning base load is 12.94 kWh/day, the air-conditioning energy of 19.80 kWh/day is scaled by a monthly utilisation factor, and a winter heating allowance is added for December to March and November.

Table 4: Monthly load model

Month	AC utilisation	Heating (kWh/day)	Daily load (kWh/day)	Monthly load (kWh)
Jan	0%	3.0	15.94	494
Feb	0%	2.0	14.94	418
Mar	10%	0.5	15.42	478
Apr	45%	0.0	21.85	656
May	100%	0.0	32.74	1,015
Jun	100%	0.0	32.74	982
Jul	90%	0.0	30.76	954
Aug	90%	0.0	30.76	954
Sep	70%	0.0	26.80	804
Oct	25%	0.0	17.89	555
Nov	0%	1.5	14.44	433
Dec	0%	3.0	15.94	494
Annual			22.56 (avg)	8,236

Table 5: Summary of consumption

Period	Consumption
Peak-summer day	32.74 kWh/day (32.74 units/day)
Peak month (May)	1,015 kWh/month
Average month	686 kWh/month
Annual	8,236 kWh/year (8,236 units/year)

5. Peak Load Estimation

The connected load is the sum of the ratings of all appliances that could run in the same hour. A diversity factor of 0.85 is applied because not every appliance runs at full power together, and a further 25% margin covers motor starting currents and future additions.

Table 6: Connected load and design demand

Appliance / load	Power (kW)
3 air conditioners (3 x 1.4 kW)	4.20
6 ceiling fans (6 x 0.075 kW)	0.45
Refrigerator	0.25
Water pump	0.75
LED lighting (20 x 12 W)	0.24
TV and set-top boxes	0.20
Washing machine	0.50
Kitchen appliances	1.20
Wi-Fi, laptops, chargers	0.25
Total connected load	8.04
Coincident demand (diversity factor 0.85)	6.83
Design demand (+25% margin)	8.54

Connected load = $4.20 + 0.45 + 0.25 + 0.75 + 0.24 + 0.20 + 0.50 + 1.20 + 0.25 = 8.04$ kW

Coincident demand = $8.04 \times 0.85 = 6.83$ kW

Design demand = $6.83 \times 1.25 = 8.54$ kW ≈ 8.5 kW

A sanctioned load of at least 8 kW is therefore appropriate. This matters for the grid connection, because the rooftop capacity PSPCL permits is tied to the consumer's sanctioned load; the 6.6 kWp array selected later in this report sits within it.

6. Essential Load for Battery Backup

Backing up the whole house for seven days would need an impractically large battery, so the backup is limited to the loads that must keep running during an outage. The air conditioners, kitchen appliances and washing machine are excluded. During an outage the household is assumed to run a reduced pattern of use: four of the six fans, one pumping cycle a day and fewer hours of lighting and TV.

Table 7: Essential load during grid failure

Essential load	Quantity	Power (kW)	Hours/day	Energy (kWh/day)
Ceiling fans	4 of 6 fans	0.075	10.0	3.00
Refrigerator	1 unit	0.20	9.0	1.80
Water pump (restricted use)	1 unit	0.75	1.0	0.75
LED lighting	12 lamps	0.012	5.0	0.72
TV and set-top box	1 set	0.10	3.0	0.30

Essential load	Quantity	Power (kW)	Hours/day	Energy (kWh/day)
Wi-Fi router, laptops, chargers	Lumped	0.20	8.0	1.60
Total essential load				8.17

Essential load = $3.00 + 1.80 + 0.75 + 0.72 + 0.30 + 1.60 = 8.17$ kWh/day

Backup energy for 7 days = $8.17 \times 7 = 57.19$ kWh

The essential loads draw about 1.75 kW when running together (fans 0.30 kW, refrigerator 0.25 kW, pump 0.75 kW, lighting 0.14 kW, TV 0.10 kW, IT 0.20 kW). Allowing for the starting surge of the pump and refrigerator compressor, the peak is about 3.8 kW, which is within the 6 kW backup rating of the selected inverter. The essential load is 25% of the full-house load.

7. Battery Bank Sizing (7-Day Autonomy)

7.1 Battery Technology

LiFePO₄ (lithium iron phosphate) is selected over lead-acid for the following reasons:

- Usable depth of discharge of 90% or more, against about 50% for lead-acid.
- Cycle life of 3,000 to 6,000 cycles or more, and a calendar life of about 10 to 15 years.
- Round-trip efficiency above 90%, and almost no maintenance.
- The most thermally stable lithium chemistry, with no gas emission or acid handling.
- Direct communication with modern hybrid inverters through a CAN or RS485 battery-management link.

7.2 Battery Capacity Calculation

Three factors reduce the nameplate energy that is actually available to the load. They are the usable depth of discharge (DoD), the discharge-path efficiency (battery discharge and inverter DC-to-AC conversion), and a derating for ageing and low-temperature operation so that the 7-day target is still met late in the battery's life.

Table 8: Battery design factors

Factor	Value	Basis
Usable depth of discharge (DoD)	90%	Typical for LiFePO ₄ rack modules
Discharge-path efficiency	92%	About 96% battery discharge $\times$ 96% inverter
Ageing and temperature derating	90%	Capacity retained at end of design life
Backup duration	7 days	Assignment requirement

Battery capacity = Backup energy / (DoD $\times$ Efficiency $\times$ Derating)

Battery capacity = $57.19 / (0.90 \times 0.92 \times 0.90) = 57.19 / 0.7452 = 76.74$ kWh

Using 5.12 kWh LiFePO₄ rack modules (51.2 V, 100 Ah), the number of modules is $76.74 / 5.12 = 14.99$, rounded up to 15.

Selected battery bank = 15×5.12 kWh = 76.8 kWh LiFePO₄

Usable energy (new battery) = $76.8 \times 0.90 \times 0.92 = 63.59$ kWh

Backup duration (new battery) = $63.59 / 8.17 = 7.78$ days

Usable energy at 90% capacity = $63.59 \times 0.90 = 57.23$ kWh, giving 7.01 days

The 76.8 kWh bank therefore delivers about 7.8 days of essential-load backup when new and still meets the full 7 days after it has aged to 90% of its rated capacity. The average essential load of

about 0.34 kW is under 0.5% of the bank's capacity per hour, so the battery works at a very gentle rate.

7.3 Recharging After an Outage

Over 7 days the battery delivers 57.19 kWh to the loads, which is 62.2 kWh at the battery terminals, and about 65.4 kWh must be put back allowing 95% charging efficiency. Once the grid returns, the inverter can recharge the battery from the grid and PV together.

Grid recharge time at 5 kW = $65.4 / 5 = 13.1$ hours

Recharging from the daily solar surplus alone is much slower and depends on the season. Using the monthly balance in Section 15, the surplus would refill the battery in about 5 days in March, 7 days in October, and roughly 53 to 65 days in May and September. There is no surplus in June, July or August. The energy-management settings should therefore allow grid charging after an outage.

Note: The 7-day figure is the conservative case in which the PV array contributes nothing during the outage. In island mode the inverter also runs the essential loads directly from PV during daylight, so on ordinary days the battery discharges much less than the figure above assumes.

8. Full-House Backup: Comparison and Justification

To justify limiting the backup to essential loads, the battery needed for a full-house 7-day backup is calculated below.

Full-house energy for 7 days = $32.74 \times 7 = 229.18$ kWh
 Battery capacity = $229.18 / 0.7452 = 307.5$ kWh
 Approximate cost at ₹22,000 per kWh = ₹67.66 lakh

A bank of about 308 kWh is 4.0 times the essential-load bank, would cost roughly ₹68 lakh, and would need a space and fire-safety arrangement closer to a commercial installation than a home. Limiting the backup to essential loads is the sensible engineering decision. The air conditioners run from solar during the day and from the grid at other times, and are not backed up by the battery.

9. Solar Resource and Performance Ratio

9.1 Peak Sun Hours for Patiala

Peak sun hours (PSH) express the daily solar energy on the module plane as an equivalent number of hours at 1,000 W/m². For a module tilted at 25° and facing south, representative monthly values for Patiala are given in Table 9. These are assumed design values typical of north-western India, with lower winter values that allow for fog and haze. They should be replaced with site data from PVGIS or the Global Solar Atlas before any real installation.

Table 9: Assumed peak sun hours on a 25° tilted plane (kWh/m²/day)

Month	Jan	Feb	Mar	Apr	May	Jun
PSH	4.0	4.8	5.7	6.4	6.6	6.2
Month	Jul	Aug	Sep	Oct	Nov	Dec
PSH	5.0	4.9	5.4	5.3	4.6	3.9

The annual mean is 5.23 PSH/day, equal to 1,910 kWh/m² per year on the module plane.

9.2 Performance Ratio

The performance ratio (PR) accounts for all the losses between the irradiation received and the AC energy delivered. It is built up from the loss budget in Table 10.

Table 10: Loss budget

Loss factor	Loss	Retained fraction
Module temperature (annual average)	7.5%	0.925
Soiling (dust, fortnightly cleaning)	4.0%	0.960
Module mismatch and quality	1.5%	0.985
Angle-of-incidence and spectral	2.0%	0.980
DC wiring	1.0%	0.990
Inverter (MPPT + conversion)	2.5%	0.975
AC wiring	0.5%	0.995
Availability / grid downtime	1.0%	0.990
Computed performance ratio		0.815
Design performance ratio adopted		0.78

The design PR of 0.78 is about 4% below the computed 0.815, which leaves a margin for first-year module degradation and uncertainty in the resource data.

10. Solar PV Array Sizing

Two methods are used to bound the array size.

10.1 Method A: Peak-day energy

$$\begin{aligned} \text{PV size} &= \text{Daily energy} / (\text{PSH} \times \text{PR}) \\ \text{PV size} &= 32.74 / (5.23 \times 0.78) = 8.02 \text{ kWp} \end{aligned}$$

This sizes the array so that even a peak-summer day is met entirely from PV at average irradiation. It is the upper bound, because the house does not run at this level for most of the year.

10.2 Method B: Annual net-metering balance

PSPCL settles net metering on an annual basis, so units exported in winter and spring are banked and offset the units imported in monsoon months. The annual energy balance is therefore the governing criterion.

$$\begin{aligned} \text{PV size} &= \text{Annual energy} / (\text{Annual irradiation} \times \text{PR}) \\ \text{PV size} &= 8,236 / (1,910 \times 0.78) = 5.53 \text{ kWp} \end{aligned}$$

10.3 Selected capacity

The annual balance gives 5.5 kWp and the peak-day method gives 8.0 kWp. A capacity of **6.6 kWp** is selected, which is 19% above the annual minimum. The margin covers battery charging losses, module degradation over the years, a poor-weather year and future loads such as an EV charger or an extra air conditioner. As Section 15 shows, this size also brings generation close to the load in the peak months of May and June.

$$\text{Selected PV array capacity} = 6.6 \text{ kWp}$$

11. Solar Module Selection

The module is a 550 W monocrystalline half-cut TOPCon (Tunnel Oxide Passivated Contact) module, as required by the assignment brief. TOPCon cells offer higher efficiency, a lower temperature coefficient and better low-light response than PERC cells. The half-cut design halves the current in each cell section, reducing resistive loss and improving partial-shading tolerance.

11.1 Module electrical parameters

Table 11 lists typical datasheet values for a 550 W TOPCon module. They are assumed for this design and must be replaced with the values from the datasheet of the module actually purchased.

Table 11: 550 W module parameters (typical, at STC)

Parameter	Symbol	Value
Maximum power	P _{max}	550 W
Voltage at maximum power	V _{mp}	42.3 V
Current at maximum power	I _{mp}	13.00 A
Open-circuit voltage	V _{oc}	50.6 V
Short-circuit current	I _{sc}	13.85 A
Dimensions	L × W	2,278 × 1,134 mm (2.58 m ²)
Module efficiency	η	21.3%
Temperature coefficient of P _{max}		-0.30 %/°C
Temperature coefficient of V _{oc}		-0.25 %/°C
Temperature coefficient of V _{mp} (assumed)		-0.30 %/°C
Temperature coefficient of I _{sc}		+0.046 %/°C

11.2 Design significance of V_{mp} and V_{oc}

V_{mp} is the voltage at which the module delivers maximum power under standard test conditions (1,000 W/m², 25 °C, AM 1.5). It is used to check that the string operating voltage stays inside the MPPT window of the inverter. **V_{oc}** is the voltage with no load connected, and it is always higher than V_{mp}. V_{oc} rises in cold weather, so it is used to make sure the string never exceeds the maximum DC input voltage of the inverter.

11.3 Number of modules

Number of modules = 6,600 W / 550 W = 12
 Installed PV capacity = 12 × 550 W = 6,600 W = 6.6 kWp

12. Rooftop Layout and Feasibility

12.1 Tilt and row spacing

The modules face true south at a fixed tilt of 25°, which is close to the optimum annual angle for Patiala and keeps the structure low. The row pitch is set so that the front row does not shade the rear row at solar noon on the winter solstice, when the sun is lowest.

Sun altitude at winter solstice noon = $90^\circ - 30.34^\circ - 23.44^\circ = 36.22^\circ$

Module height above roof = $2.278 \times \sin 25^\circ = 0.963 \text{ m}$

Horizontal projection = $2.278 \times \cos 25^\circ = 2.065 \text{ m}$

Shadow length = $0.963 / \tan 36.22^\circ = 1.314 \text{ m}$

Minimum row pitch = $2.065 + 1.314 = 3.379 \text{ m}$; adopted pitch = 3.5 m

12.2 Array layout and area check

The 12 modules are mounted in portrait orientation in two rows of six. Each row forms one string. A 0.6 m walkway is left around the array for maintenance.

Array width = $6 \times 1.134 + 5 \times 0.02 = 6.90 \text{ m}$

Array depth = $2.065 + 3.5 = 5.56 \text{ m}$

Envelope with 0.6 m walkway = $8.10 \times 6.76 = 54.8 \text{ m}^2$

Module area = $12 \times 2.583 = 31.0 \text{ m}^2$; practical area factor = $54.8 / 31.0 = 1.77$

The layout-based factor of 1.77 falls within the usual 1.5 to 1.8 range for rooftop arrays, giving a practical requirement of 46 to 56 m^2 (54.8 m^2 from the detailed layout). The usable roof is 94 m^2 , so the system is feasible with about 39 m^2 to spare, enough for up to about 8 more modules if the load grows. The array is placed south of the mumty and water tanks so their shadows fall away from it.

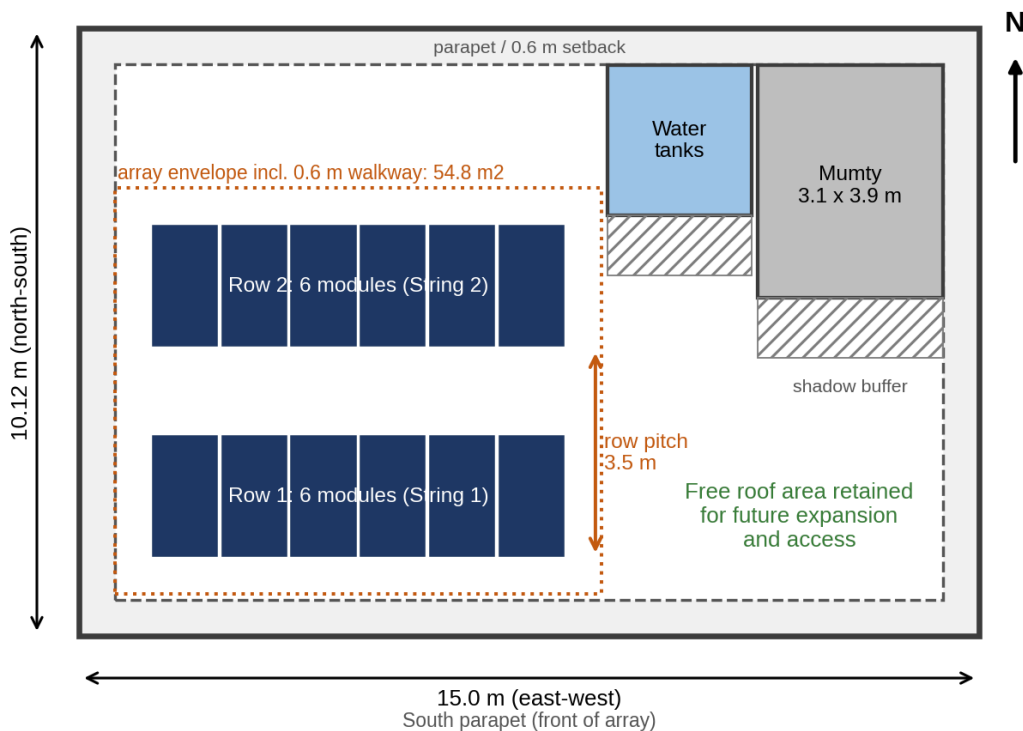


Figure 1: Indicative rooftop layout (plan view, north at top)

13. PV String Configuration

The 12 modules are arranged as **2 strings of 6 modules in series**. Each string is connected to its own MPPT input of the inverter, so the two strings are never paralleled. This gives each row independent maximum-power tracking, which helps in the early morning and late afternoon when the front row can shade the rear row. Six modules in series also keep the voltage well below the inverter maximum while staying above its minimum MPPT voltage even on the hottest days.

13.1 String voltage and current

String Vmp (STC) = $6 \times 42.3 = 253.8 \text{ V}$; String Voc (STC) = $6 \times 50.6 = 303.6 \text{ V}$
 Current per string = $I_{mp} = 13.00 \text{ A}$ (series connection: the current does not add)
 Power per string = $6 \times 550 = 3,300 \text{ W}$

13.2 Extreme-temperature checks

The STC values are not enough. On a cold winter morning Voc is higher, and on a hot afternoon Vmp is lower. Both extremes are checked against the inverter window.

Cold Voc per module at $-5^\circ\text{C} = 50.6 \times [1 + 0.0025 \times (25 - (-5))] = 54.39 \text{ V}$
 String Voc (cold) = $6 \times 54.39 = 326.4 \text{ V}$
 Hot Vmp per module at 70°C cell = $42.3 \times [1 - 0.003 \times (70 - 25)] = 36.59 \text{ V}$
 String Vmp (hot) = $6 \times 36.59 = 219.5 \text{ V}$
 Design current for cables and switches = $1.25 \times I_{sc} = 1.25 \times 13.85 = 17.3 \text{ A}$

Table 12: String design versus inverter DC input requirements

Parameter	Inverter requirement (typical)	Design value	Check
Maximum DC input voltage	$\geq 1,000 \text{ V}$	326 V (cold Voc)	Pass
MPPT voltage range	160 to 850 V	220 to 277 V	Pass
Start-up voltage	$\leq 180 \text{ V}$	220 V (hot Vmp)	Pass
Number of MPPT inputs	2	2 (one string each)	Pass
Maximum input current per MPPT	$\geq 15 \text{ A}$	13.0 A	Pass
Maximum Isc per MPPT	$\geq 20 \text{ A}$	13.85 A (17.3 A with 1.25 factor)	Pass
DC/AC ratio	1.0 to 1.3	1.10	Pass

Note: The inverter limits in Table 12 are typical for a 6 kW three-phase hybrid inverter and must be confirmed against the datasheet of the model finally selected. A production design should also use a site-specific record low temperature for the cold Voc check.

Table 13: Final string design summary

Parameter	Value
Total modules	12
String configuration	2 strings $\times$ 6 modules in series, one string per MPPT
Module rating	550 W
Total PV capacity	6.6 kWp
String Vmp / Voc at STC	253.8 V / 303.6 V
String Voc at -5°C	326.4 V
String Vmp at 70°C cell temperature	219.5 V
Current per string (I_{mp} / I_{sc})	13.00 A / 13.85 A

14. Hybrid Inverter Sizing and Selection

Selected inverter: **6 kW three-phase hybrid solar inverter**, with a 6.6 kWp array (DC/AC ratio 1.10) and a design house demand of 8.5 kW.

14.1 Justification

- A DC/AC ratio of 1.10 is in the normal 1.0 to 1.3 range, so PV clipping is negligible.
- The 6 kW backup (EPS) rating comfortably covers the 3.8 kW essential-load peak including starting surge.
- During grid operation the whole-house load is fed from the grid and inverter together, so the inverter does not have to carry the full 8.5 kW demand.
- A three-phase inverter suits a three-phase house connection and spreads the injected power across the phases. The permissible injection on the actual connection should be confirmed with PSPCL.
- It manages the PV array, the battery and the grid, and can charge the LiFePO₄ bank from PV and from the grid.

14.2 Required inverter specifications

Table 14: Inverter specification

Specification	Requirement
Type	Three-phase hybrid solar inverter, grid-tied with battery port and backup output
AC output	415 V three-phase, 50 Hz; supports unbalanced load on the backup output
Rated AC power	6 kW
PV inputs	2 independent MPPTs; 1,000 V maximum; window as in Table 12
Battery	High-voltage LiFePO ₄ port with CAN/RS485 BMS communication; charge from PV and grid
Grid functions	Import, export and net-metering support; adjustable export limit
Anti-islanding	Mandatory, to IEC 62116 (disconnection within 2 s of grid loss)
Protections	Over/under voltage and frequency, insulation and earth-fault monitoring, DC reverse polarity, surge protection
Backup changeover	Automatic, ≤ 20 ms, to essential-load panel
Enclosure and efficiency	IP65 or better; European efficiency of 97% or more

Design caution: A 76.8 kWh battery is larger than many residential hybrid inverters can accept on their battery port. The chosen inverter must support this capacity through parallel battery clusters, or a second battery inverter must be added. This has to be confirmed with the manufacturer before ordering.

15. Solar Generation and Monthly Energy Balance

Generation = PV capacity × PSH × PR

Average daily generation = $6.6 \times 5.23 \times 0.78 = 26.9$ kWh/day

Annual generation = $6.6 \times 1,910 \times 0.78 = 9,832$ kWh/year (specific yield 1,490 kWh/kWp)

Table 15: Monthly energy balance

Month	PSH	PV (kWh/day)	Load (kWh)	PV (kWh)	Net (kWh)
Jan	4.0	20.6	494	638	+144
Feb	4.8	24.7	418	692	+274
Mar	5.7	29.3	478	910	+432
Apr	6.4	32.9	656	988	+333

Month	PSH	PV (kWh/day)	Load (kWh)	PV (kWh)	Net (kWh)
May	6.6	34.0	1,015	1,053	+38
Jun	6.2	31.9	982	958	-25
Jul	5.0	25.7	954	798	-156
Aug	4.9	25.2	954	782	-172
Sep	5.4	27.8	804	834	+30
Oct	5.3	27.3	555	846	+291
Nov	4.6	23.7	433	710	+277
Dec	3.9	20.1	494	622	+128
Annual	5.23	26.9	8,236	9,832	+1,595

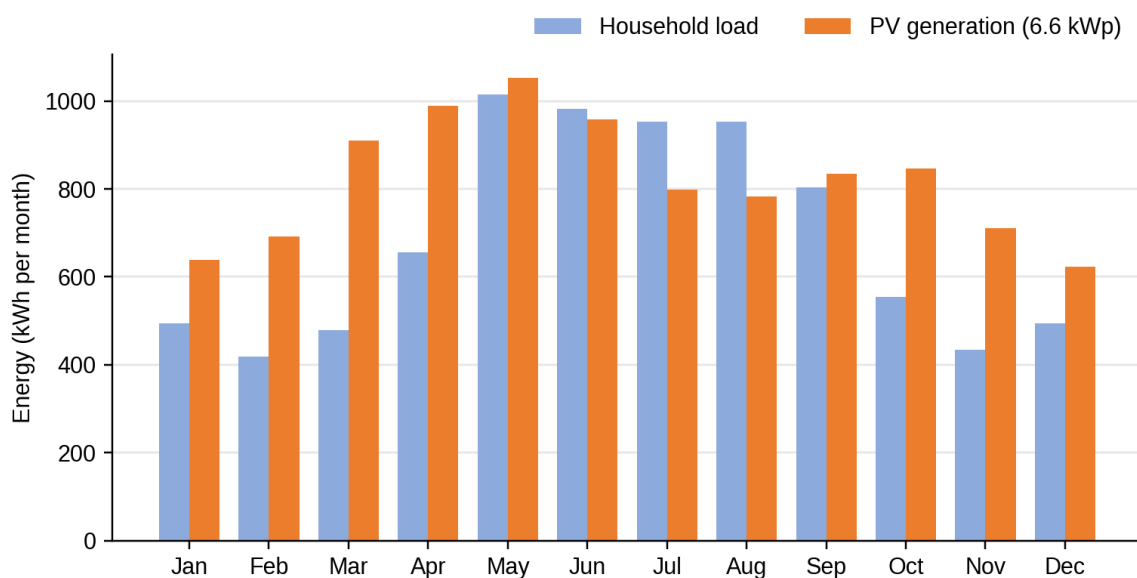


Figure 2: Monthly household load and PV generation

Annual generation of 9,832 kWh covers 119% of the 8,236 kWh annual load, leaving a net surplus of about 1,595 kWh. Generation is very close to the load in May (1,053 kWh against 1,015 kWh). The array falls short in June, July and August (by 25, 156 and 172 kWh) because of monsoon cloud and air conditioning, but these deficits are covered by units banked earlier in the year under annual net metering.

These are theoretical averages. Actual values will vary with cloud and haze, dust, shading, module temperature (output drops about 0.30% per °C above 25 °C), battery and inverter losses, and the actual household load pattern.

16. Cable Sizing and Protection Devices

16.1 DC side (string cables)

Design current = $1.25 \times I_{sc} = 1.25 \times 13.85 = 17.3 \text{ A}$

Cable: 6 mm² solar-rated copper cable (UV-resistant, double-insulated, to EN 50618 / IEC 62930)

Voltage-drop check for a 25 m one-way run (50 m loop), $\rho(\text{Cu}, 70^\circ\text{C}) = 0.021 \Omega \cdot \text{mm}^2/\text{m}$:

$R = 0.021 \times 50 / 6 = 0.175 \Omega$; drop = $13.0 \text{ A} \times 0.175 \Omega = 2.28 \text{ V} = 0.90\%$ of string V_{mp} (limit 1%)

A 4 mm² cable would drop about 1.3% over the same run, which is why 6 mm² is chosen. The values are indicative; the final size depends on the actual run length, routing and grouping.

16.2 AC side

Inverter AC current = $P / (\sqrt{3} \times V) = 6,000 / (1.732 \times 415) = 8.35 \text{ A}$

With 1.25 factor = 10.43 A, so a 16 A 4-pole MCB is selected

Cable: 4-core 4 mm² armoured copper; for a 30 m run, $R = 0.021 \times 30 / 4 = 0.158 \Omega$ per conductor

Voltage drop = $\sqrt{3} \times 8.35 \times 0.158 = 2.28 \text{ V} = 0.55\%$ of 415 V (limit 1%)

16.3 Protection devices

Table 16: Protection schedule

Section	Device	Rating / note
DC side		
	DC isolator	1,000 V DC, 25 A ($\geq 1.25 \times I_{sc}$)
	String fuse (if fitted)	25 A gPV; between 1.5 and $2.4 \times I_{sc}$ and within the module's series-fuse rating
	Surge protection	Type II DC SPD, $U_c \geq 1.2 \times \text{cold } V_{oc}$, at the DCDB and inverter input
AC side		
	Lockable AC isolator	Accessible to PSPCL for maintenance isolation
	MCB	4-pole, 16 A, C-curve
	RCCB	4-pole, 30 mA; type A or B as the inverter manufacturer requires
	Surge protection	Type II AC SPD at the distribution board
	Meter	Bidirectional net meter to PSPCL specification
Battery side		
	BMS	Monitors state of charge, cell voltage, temperature and overcurrent; CAN/RS485 link to inverter
	DC breaker and fuse	Sized to the inverter's maximum battery current, per manufacturer
	Enclosure	Ventilated, fire-rated, IEC 62619-certified modules, smoke and heat detection
General		
	Earthing	Module frames, structure and equipment bonded to low-resistance earth pits (target below 5 Ω), IS 3043
	Lightning protection	Air terminal and down-conductor bonded to the earth system (IS/IEC 62305)

Note: Cable and protection ratings are indicative. Final sizing must be confirmed by a licensed electrical engineer for the actual cable lengths, conduit type, ambient temperature, grouping factors and local standards such as IS 694 and IS 1554.

17. Grid Synchronisation, Anti-Islanding and Net Metering

17.1 Grid synchronisation

The hybrid inverter turns DC from the PV array and battery into AC. To run in parallel with the grid it must match the grid in three respects:

- **Voltage:** 230 V line-to-neutral (415 V line-to-line), within the permitted window.
- **Frequency:** 50 Hz, the Indian grid frequency.
- **Phase angle and sequence:** the output waveform must be aligned with the grid waveform on all three phases.

A phase-locked loop (PLL) inside the inverter continuously tracks grid voltage, frequency and phase. After a grid outage the inverter reconnects only when the grid has stayed within limits for a set delay, typically one to five minutes.

17.2 Power flow under different conditions

Table 17: Power flow

Condition	Power flow
Solar equals load	PV supplies the full house load. No grid import and no battery exchange.
Solar is less than load	PV and the grid together supply the load. The battery stays on reserve.
Solar exceeds load, battery not full	PV supplies the load, the surplus charges the battery, and any remainder is exported.
Solar exceeds load, battery full	PV supplies the load and all surplus is exported through the net meter.
Night, grid available	The grid supplies the load. The battery holds its backup reserve for outages.
Grid failure	The inverter isolates from the grid and supplies the essential-load panel from the battery, with PV support in daylight. Air conditioners and other non-essential loads are disconnected.
Grid restored after outage	The inverter resynchronises, reconnects, and recharges the battery from the grid and PV.

Because the battery exists mainly for outages, its backup-reserve setting should be kept high so that it is not drained by everyday use. Daily cycling would save little under the slab tariff used here, where no time-of-day price difference is assumed.

17.3 Anti-islanding protection

Anti-islanding is a mandatory safety function for every grid-connected inverter. If the grid fails, the inverter must disconnect within a short time limit (2 seconds under the IEC 62116 test). Otherwise it would keep energising a dead network and endanger utility staff working on it. Detection relies on over/under-voltage and over/under-frequency limits, rate-of-change-of-frequency, and active islanding-detection methods. In this hybrid system the inverter isolates from the grid and then forms a local island that supplies only the essential-load panel. The lockable AC isolator gives PSPCL a visible point of isolation.

17.4 Net metering under PSPCL

Under the PSERC rooftop solar regulations, a consumer may install a grid-interactive system with net metering. A bidirectional meter records imports and exports. The billed energy is the net of the two, and any excess is banked and settled on an annual basis. The application is made through the

PSPCL solar portal, and technical feasibility depends on the consumer's sanctioned load and the capacity of the local distribution transformer. Electricity duty is levied only on the net billed amount.

18. System Block Diagram

Figure 3 shows the architecture of the proposed hybrid system and the direction of power flow between its main parts.

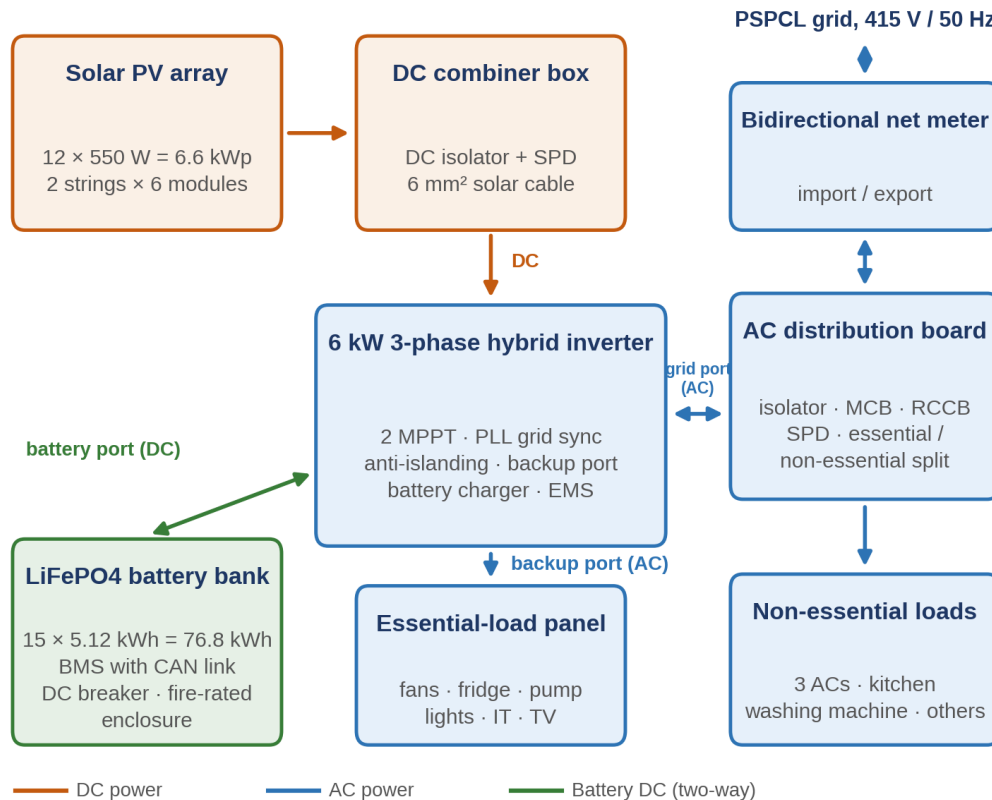


Figure 3: Block diagram of the grid-connected hybrid solar PV system

18.1 Component description

- **Solar PV array:** 12 modules of 550 W (6.6 kWp) in two series strings of six modules.
- **DC combiner box:** carries the DC isolator and surge protection, and passes each string to its own MPPT input.
- **Hybrid inverter (6 kW, three-phase):** converts DC to AC, tracks the maximum power point, manages flow between PV, battery, loads and grid, and provides the backup output.
- **Battery bank:** 15 × 5.12 kWh LiFePO4 modules (76.8 kWh) with a BMS that talks to the inverter.
- **AC distribution board:** feeds the household circuits and separates essential from non-essential loads.
- **Bidirectional net meter:** records energy imported from and exported to the PSPCL grid.

19. Cost Estimate

The figures below are indicative 2026 Indian market values and are meant for academic estimation. They are based on retail price listings for LFP batteries (roughly ₹20,000 to ₹28,000 per kWh) and

hybrid inverters, and actual costs will vary with brand, supplier and site. Final figures should come from installer quotations.

Table 18: Component-wise cost of the hybrid system

Component	Basis	Cost (₹)
PV modules	12 × 550 W TOPCon at ₹25/Wp	1,65,000
Mounting structure	6.6 kWp at ₹5,000/kWp, hot-dip galvanised	33,000
Hybrid inverter	6 kW three-phase, lump sum	1,20,000
LiFePO4 battery bank	76.8 kWh at ₹22,000/kWh	16,89,600
DC-side balance of system	Cables, DCDB, isolator, SPD	24,000
AC-side balance of system	Cable, ACDB, MCB, RCCB, SPD, essential panel	36,000
Earthing and lightning protection	Lump sum	18,000
Battery enclosure and safety	Rack, breaker, ventilation, fire protection	45,000
Net metering	Application, bidirectional meter, drawings	15,000
Installation and commissioning	Labour, transport, civil work	60,000
Total estimated project cost		22,05,600 (₹22.06 lakh)

The battery bank costs ₹16.90 lakh, which is 77% of the total project cost. It dominates the budget, and this is a direct consequence of specifying seven days of autonomy.

Table 19: Solar-only comparison system (PV with on-grid inverter, no battery)

Component	Cost (₹)
PV modules and mounting structure	1,98,000
6 kW three-phase on-grid inverter	70,000
DC and AC balance of system, earthing	72,000
Net metering, installation and commissioning	60,000
Total solar-only cost	4,00,000 (₹4.00 lakh)

20. Electricity Savings and Payback Period

20.1 Annual savings

The PSERC tariff order for FY 2026-27 sets the domestic energy charge at ₹3.85/kWh for the first 300 units in a month and ₹7.05/kWh above that. This household exceeds 300 units in every month of the load model, so each month is billed at both slab rates. Because the array generates more than the household uses over the year, net annual import is close to zero and the saving is taken as the annual energy-charge bill that would otherwise be paid.

Table 20: Monthly energy-charge bill without solar

Month	Jan	Feb	Mar	Apr	May	Jun
Bill (₹)	2,524	1,989	2,410	3,661	6,195	5,965
Month	Jul	Aug	Sep	Oct	Nov	Dec

Bill (₹)	5,763	5,763	4,708	2,950	2,094	2,524
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Monthly bill = $300 \times ₹3.85 + (\text{Units} - 300) \times ₹7.05$ (for example, May: $300 \times 3.85 + 715 \times 7.05 = ₹6,195$)

Annual saving = ₹46,545 (about ₹0.47 lakh per year)

20.2 Simple payback

Table 21: Payback comparison

Scenario	System cost	Annual saving	Simple payback
Hybrid system with 76.8 kWh battery	₹22.06 lakh	₹0.47 lakh	47.4 years
Hybrid system, after ₹78,000 subsidy	₹21.28 lakh	₹0.47 lakh	45.7 years
Solar-only system (no battery)	₹4.00 lakh	₹0.47 lakh	8.6 years
Solar-only, after ₹78,000 subsidy	₹3.22 lakh	₹0.47 lakh	6.9 years

The subsidy rows assume the household qualifies for the central financial assistance of up to ₹78,000 under PM Surya Ghar: Muft Bijli Yojana, which does not cover the battery and requires listed modules. Eligibility and the amount should be checked against the current scheme rules.

The 7-day battery adds ₹18.06 lakh to a ₹4.00 lakh solar system. Spread over a 12-year battery life, that is about ₹1.50 lakh per year, roughly 3.2 times the annual electricity saving. A payback of 47 years is also far longer than the battery's own life, so the battery cannot be justified as a money-saving item. It is a resilience purchase, and its value lies in keeping essential loads running through a long outage.

Assumptions: Savings use energy charges only. Electricity duty, fixed charges and fuel adjustments are excluded, PSPCL bills in kWh but unity power factor is assumed, the state free-units scheme is not modelled, and the 1,595 kWh year-end surplus is given no value. PV degradation and tariff escalation are neglected.

21. Design Risks and Recommendations

21.1 Right-sizing the battery in stages

Because the battery is modular, it can be added in stages. Table 22 shows the battery size and cost for shorter autonomy periods, using the same design factors as Section 7.

Table 22: Battery size and cost versus autonomy

Autonomy	Required (kWh)	Modules	Installed (kWh)	Battery cost (₹ lakh)
1 day	11.0	3	15.4	3.38
2 days	21.9	5	25.6	5.63
3 days	32.9	7	35.8	7.88
7 days	76.7	15	76.8	16.90

Grid outages in urban areas typically last hours rather than days, so a smaller battery would cover the common cases at a fraction of the cost. The 7-day design is kept here because the brief requires it, but the inverter and enclosure can be chosen so that modules are added over time.

21.2 Other points to address before installation

- **Inverter and battery compatibility:** confirm that the inverter accepts a 76.8 kWh bank (parallel clusters or a second battery inverter).
- **Battery safety:** house the bank in a ventilated, fire-separated space away from habitable rooms and exits, with smoke detection, and check the requirement with the local fire authority.
- **Resource data:** replace the assumed peak-sun-hours in Table 9 with PVGIS or Global Solar Atlas values for the exact site.
- **Sanctioned load and transformer capacity:** confirm the sanctioned load (8 kW or more) and the distribution transformer hosting capacity with PSPCL before applying.
- **Structure and roof:** design the mounting for local wind loads (IS 875 Part 3), protect the waterproofing at anchor points, and keep the array clear of the mumty and tank shadows.
- **Cleaning:** wash the modules about every two weeks in the dry season, since soiling is the largest avoidable loss after temperature.

22. Final Design Summary

Table 23: Final design summary

Parameter	Final design value
Site and load	
Location	Patiala, Punjab, India
System type	Grid-connected hybrid rooftop solar PV system
Peak-summer daily load	32.74 kWh/day
Annual load	8,236 kWh/year
Design demand	8.5 kW (connected load 8.04 kW)
Available / required roof area	94 m ² / 54.8 m ²
PV array	
PV module	550 W monocrystalline TOPCon half-cut
Number of modules and capacity	12 modules, 6.6 kWp
Tilt, orientation, row pitch	25°, true south, 3.5 m
String configuration	2 strings × 6 modules in series
String V _{mp} / V _{oc} (STC)	253.8 V / 303.6 V
String V _{oc} at -5 °C	326.4 V
Annual generation	9,832 kWh/year (1,490 kWh/kWp)
Inverter and battery	
Inverter	6 kW three-phase hybrid, 2 MPPT
Essential load	8.17 kWh/day
Battery	15 × 5.12 kWh LiFePO ₄ = 76.8 kWh
Backup duration	7 days (7.8 days new, 7.0 days at 90% capacity)
Electrical and economics	
DC string cable	6 mm ² solar cable
AC output cable / protection	4C × 4 mm ² Cu; 4-pole 16 A MCB, 30 mA RCCB
Total project cost	₹22.06 lakh
Annual electricity saving	₹0.47 lakh/year
Payback (with battery / solar-only)	47 years / 8.6 years

23. Conclusion

The proposed system is a 6.6 kWp grid-connected hybrid rooftop PV system for a five-member household in Patiala, Punjab. It uses 12 modules of 550 W in two strings of six, feeding a 6 kW three-phase hybrid inverter. The array is expected to generate about 9,832 kWh a year against an annual load of about 8,236 kWh, so net annual import is close to zero under PSPCL's annual net metering.

The battery is a 76.8 kWh LiFePO₄ bank sized for seven days of essential-load backup (8.17 kWh/day for fans, refrigerator, pump, lighting and IT). It provides about 7.8 days when new and still meets seven days at 90% of rated capacity. The air conditioners are deliberately left off the backup, since backing up the full house would need about 308 kWh of storage.

The site is suitable. The array needs about 55 m² of the 94 m² of usable roof, and the string voltages stay within the inverter window at both -5 °C and 70 °C cell temperature. The estimated cost is about ₹22.06 lakh, of which the battery is 77%. The solar-only part costs about ₹4.00 lakh and pays back in roughly 8.6 years.

Final Engineering Verdict

A 6.6 kWp hybrid rooftop PV system with a 76.8 kWh LiFePO₄ battery is technically feasible for the selected house. It meets the annual energy demand, provides at least seven days of essential-load backup, reduces grid dependence and allows import and export through net metering. Financially, however, the PV part is attractive and the battery is not: it multiplies the project cost more than five times and is not recovered through savings within its life. The battery should be treated as a resilience investment. The recommended approach is to install the PV array and a hybrid-ready inverter first, and to add battery modules in stages, sized to the outage duration the household actually needs to ride through.

References

- [1] Punjab State Electricity Regulatory Commission (PSERC), Tariff Order for PSPCL and PSTCL, FY 2026-27 (effective 1 April 2026 to 31 March 2027), as reported in the press in March and July 2026.
- [2] PSERC, Grid Interactive Rooftop Solar Photovoltaic Systems Regulations, 2021, as amended; PEDAs and PSPCL rooftop solar information pages.
- [3] Ministry of New and Renewable Energy, PM Surya Ghar: Muft Bijli Yojana, central financial assistance for residential rooftop solar.
- [4] European Commission JRC, PVGIS; World Bank Group and Solargis, Global Solar Atlas (site resource data, to be used to verify Table 9).
- [5] IEC 62548 (PV array design), IEC 62109-1 and -2 (inverter safety), IEC 62116 (anti-islanding), IEC 61215 and IEC 61730 (module qualification and safety, IS 14286), IEC 62619 (lithium battery safety).
- [6] IS 3043 (earthing), IS/IEC 62305 (lightning protection), IS 875 Part 3 (wind loads), IS 694 and IS 1554 (cables).
- [7] Central Electricity Authority, Technical Standards for Connectivity of the Distributed Generation Resources Regulations, as amended.
- [8] Indicative 2026 Indian retail price listings for LFP storage batteries and hybrid inverters (used for the cost estimate in Section 19).